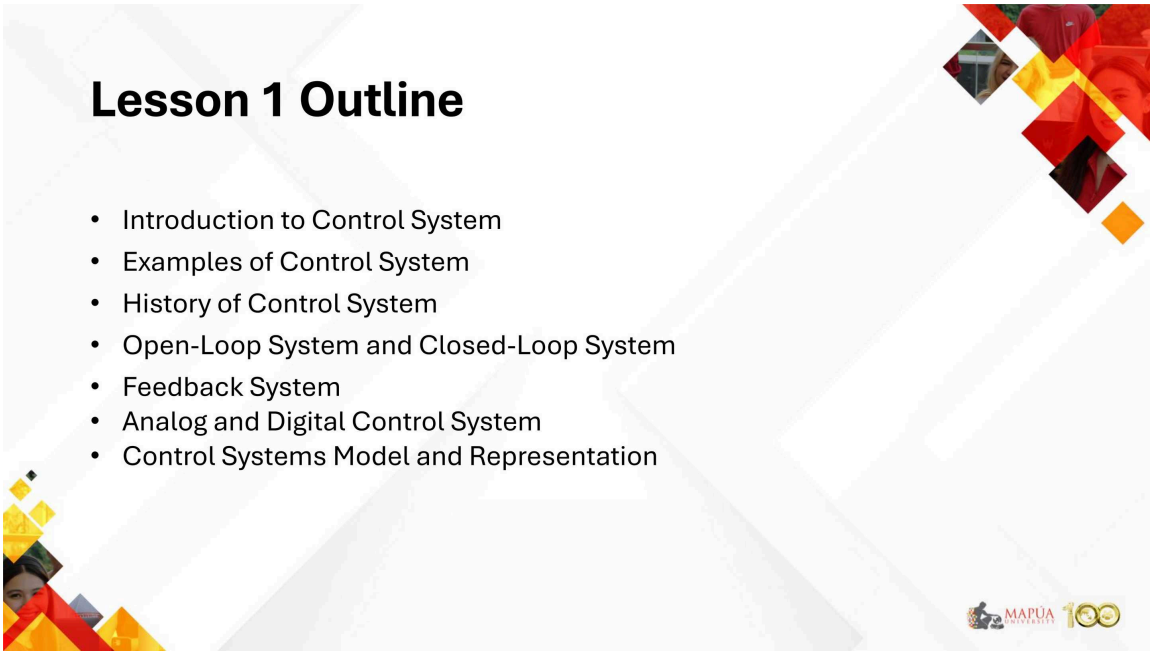


Feedback and Control Systems

Course Outcome 1

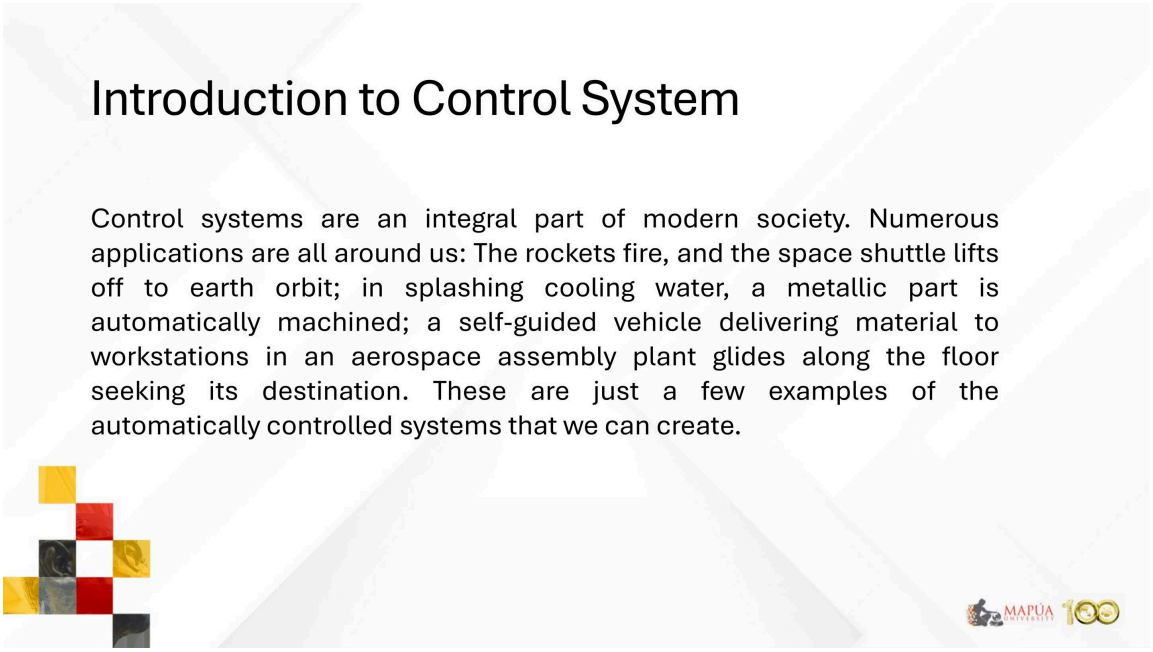
Lesson 1: Introduction to Feedback and Control Systems

By: Carlos HortinelaIV



Lesson 1 Outline

- Introduction to Control System
- Examples of Control System
- History of Control System
- Open-Loop System and Closed-Loop System
- Feedback System
- Analog and Digital Control System
- Control Systems Model and Representation



Introduction to Control System

Control systems are an integral part of modern society. Numerous applications are all around us: The rockets fire, and the space shuttle lifts off to earth orbit; in splashing cooling water, a metallic part is automatically machined; a self-guided vehicle delivering material to workstations in an aerospace assembly plant glides along the floor seeking its destination. These are just a few examples of the automatically controlled systems that we can create.

Introduction to Control System

In modern usage the word *system* has many meanings.
Definition 1.1a: A **system** is an arrangement, set, or collection of things connected or related in such a manner as to form an entirety or whole.
Definition 1.1b: A **system** is an arrangement of physical components connected or related in such a manner as to form and/or act as an entire unit.
The word **control** is usually taken to mean *regulate, direct,* or *command*.

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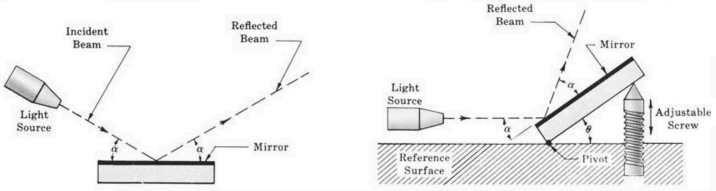
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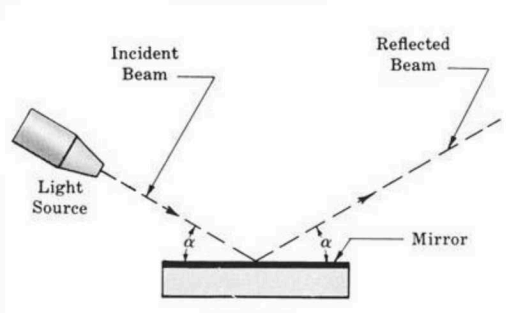
Introduction to Control System

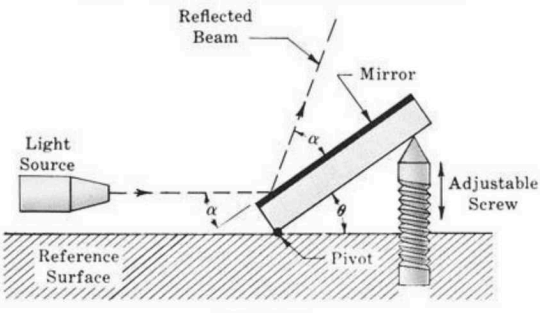
Definition 1.2a: A **control system** is an arrangement of physical components connected or related in such a manner as to command, direct, or regulate itself or another system.



Introduction to Control System

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Introduction to Control System

It is important to note, however, that control systems of interest for analysis or design purposes include not only those manufactured by humans, but those that normally exist in nature, and control systems with both manufactured and natural components.

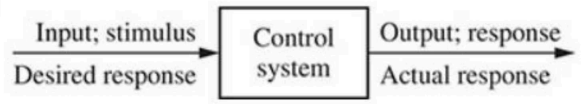


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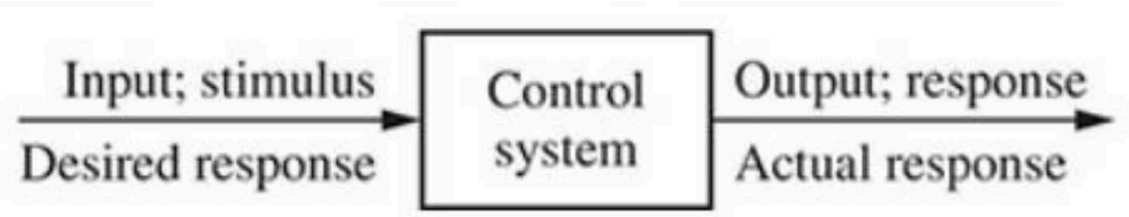
Introduction to Control System

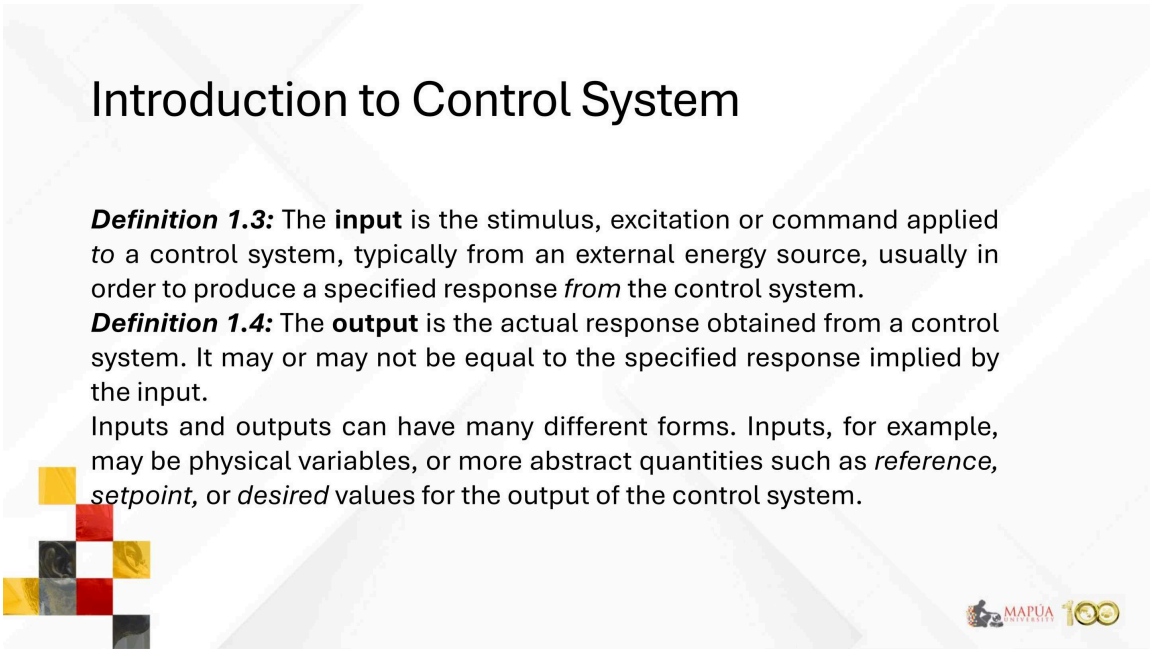
Definition 1.2b: A **control system** consists of subsystems and processes (or plants) assembled for the purpose of obtaining a desired output with desired performance, given a specified input. Figure below shows a control system in its simplest form, where the input represents a desired output.



Introduction to Control System

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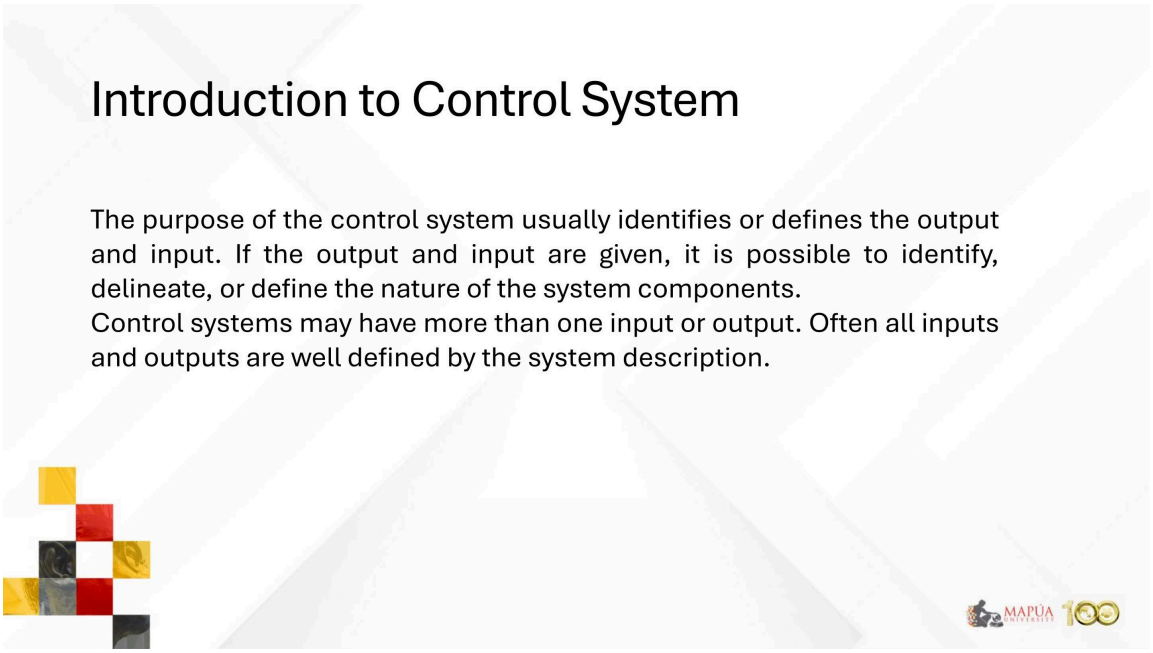


Introduction to Control System

Definition 1.3: The input is the stimulus, excitation or command applied to a control system, typically from an external energy source, usually in order to produce a specified response from the control system.

Definition 1.4: The output is the actual response obtained from a control system. It may or may not be equal to the specified response implied by the input.

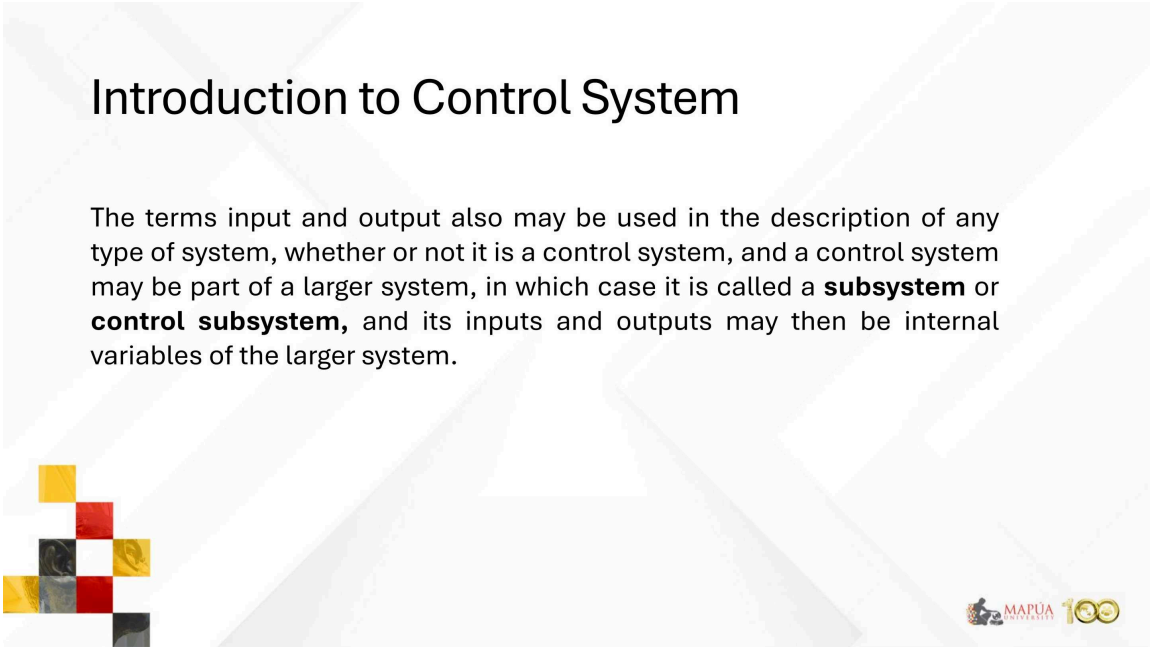
Inputs and outputs can have many different forms. Inputs, for example, may be physical variables, or more abstract quantities such as reference, setpoint, or desired values for the output of the control system.



Introduction to Control System

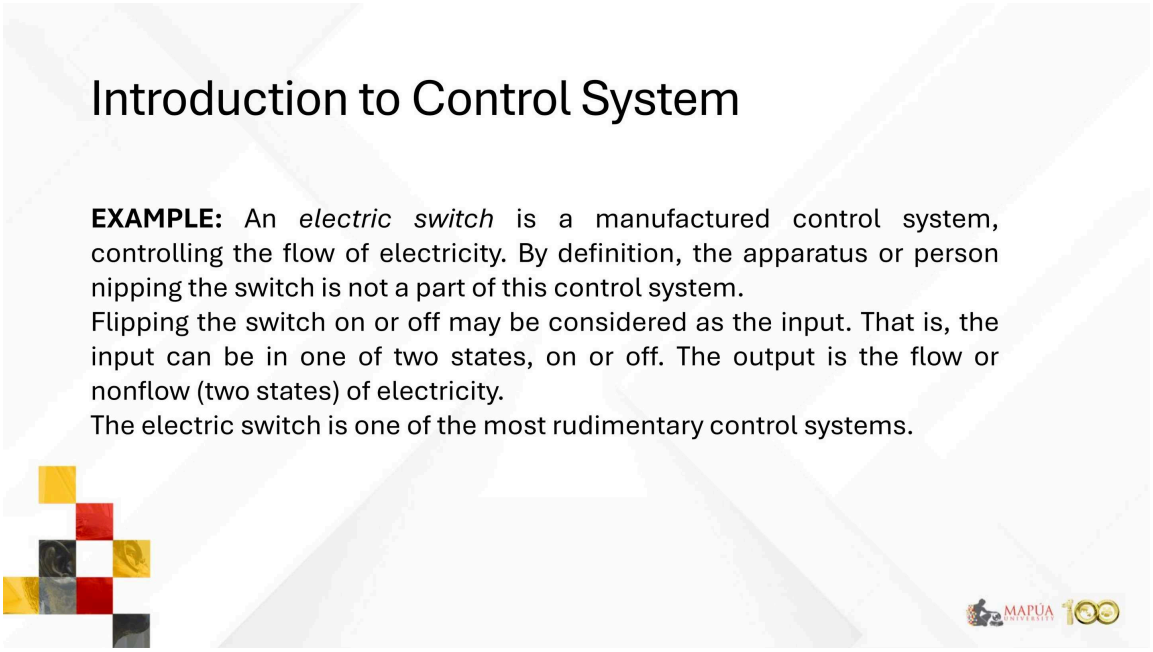
The purpose of the control system usually identifies or defines the output and input. If the output and input are given, it is possible to identify, delineate, or define the nature of the system components.

Control systems may have more than one input or output. Often all inputs and outputs are well defined by the system description.



Introduction to Control System

The terms input and output also may be used in the description of any type of system, whether or not it is a control system, and a control system may be part of a larger system, in which case it is called a subsystem or control subsystem, and its inputs and outputs may then be internal variables of the larger system.

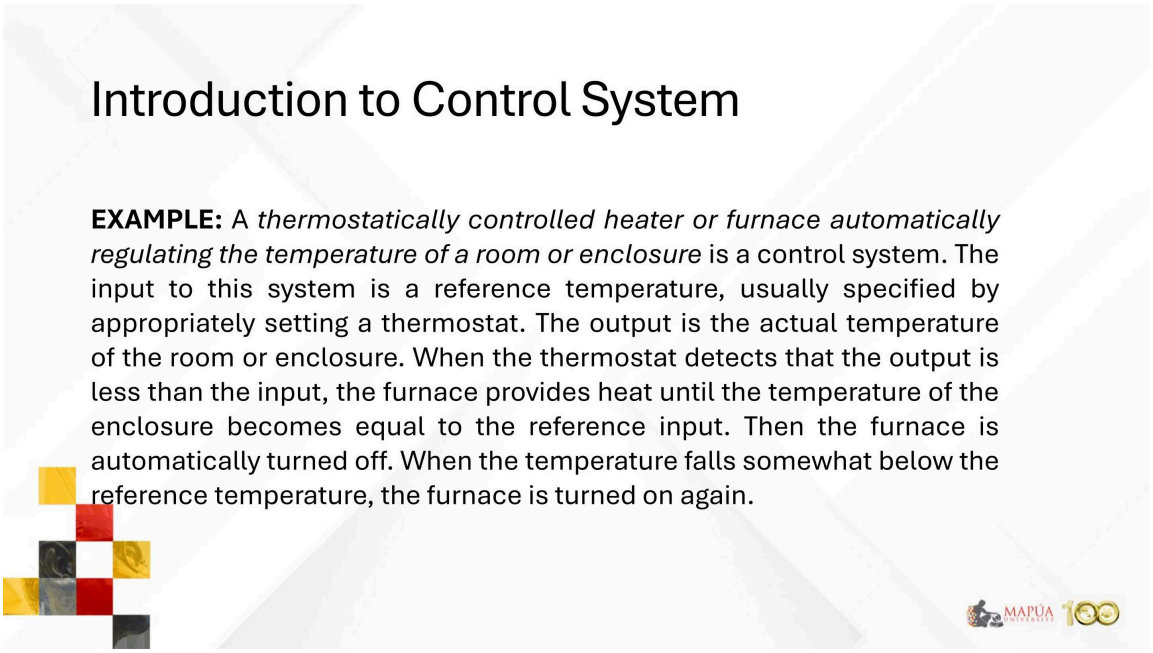


Introduction to Control System

EXAMPLE: An electric switch is a manufactured control system, controlling the flow of electricity. By definition, the apparatus or person nipping the switch is not a part of this control system.

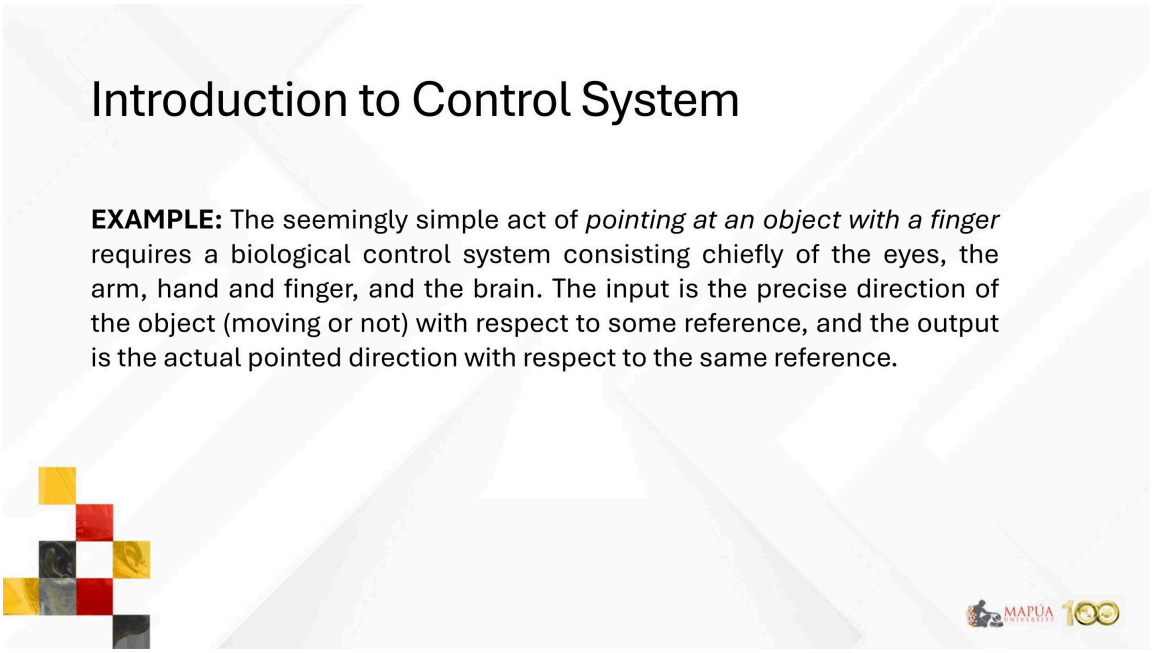
Flipping the switch on or off may be considered as the input. That is, the input can be in one of two states, on or off. The output is the flow or nonflow (two states) of electricity.

The electric switch is one of the most rudimentary control systems.



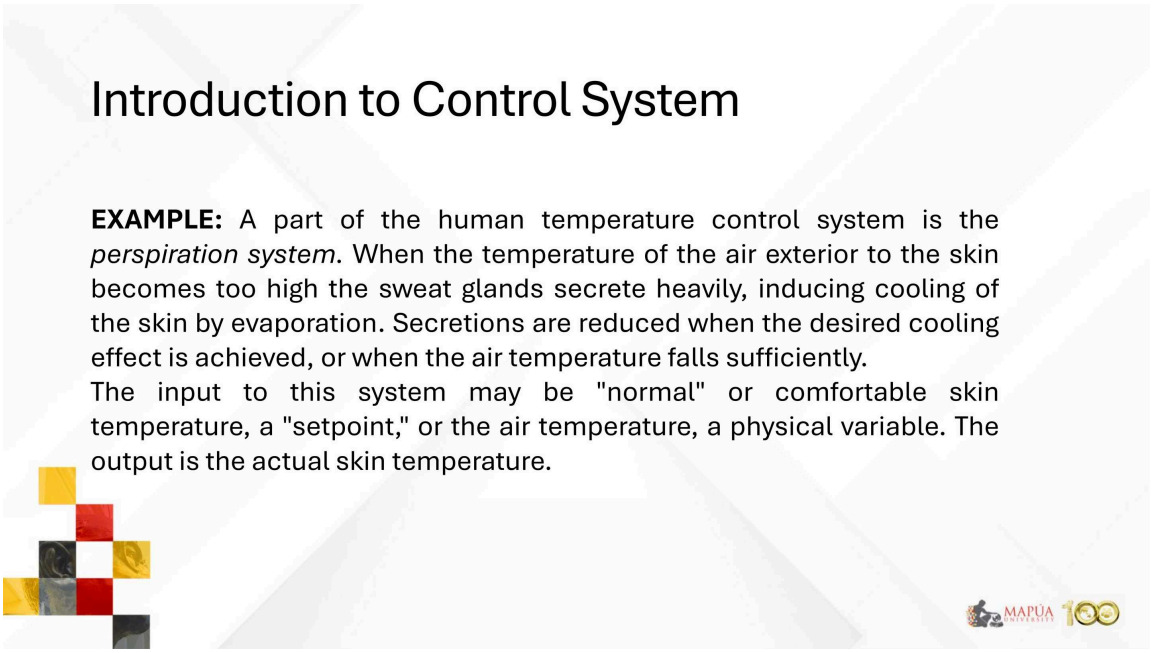
Introduction to Control System

EXAMPLE: A thermostatically controlled heater or furnace automatically regulating the temperature of a room or enclosure is a control system. The input to this system is a reference temperature, usually specified by appropriately setting a thermostat. The output is the actual temperature of the room or enclosure. When the thermostat detects that the output is less than the input, the furnace provides heat until the temperature of the enclosure becomes equal to the reference input. Then the furnace is automatically turned off. When the temperature falls somewhat below the reference temperature, the furnace is turned on again.



Introduction to Control System

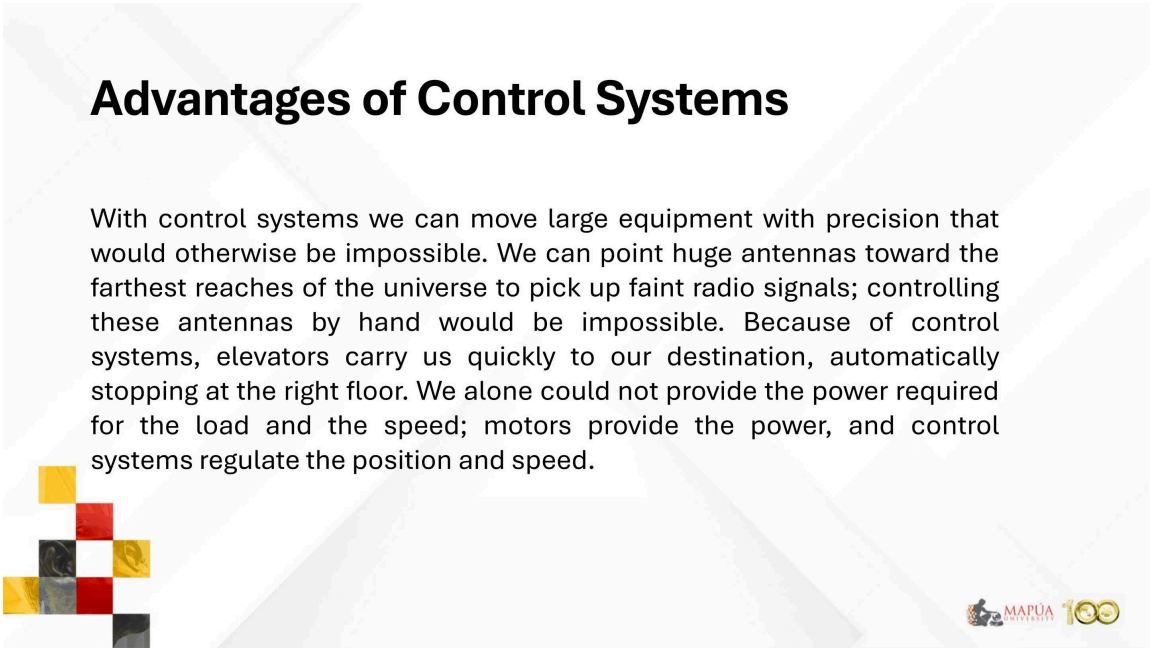
EXAMPLE: The seemingly simple act of pointing at an object with a finger requires a biological control system consisting chiefly of the eyes, the arm, hand and finger, and the brain. The input is the precise direction of the object (moving or not) with respect to some reference, and the output is the actual pointed direction with respect to the same reference.



Introduction to Control System

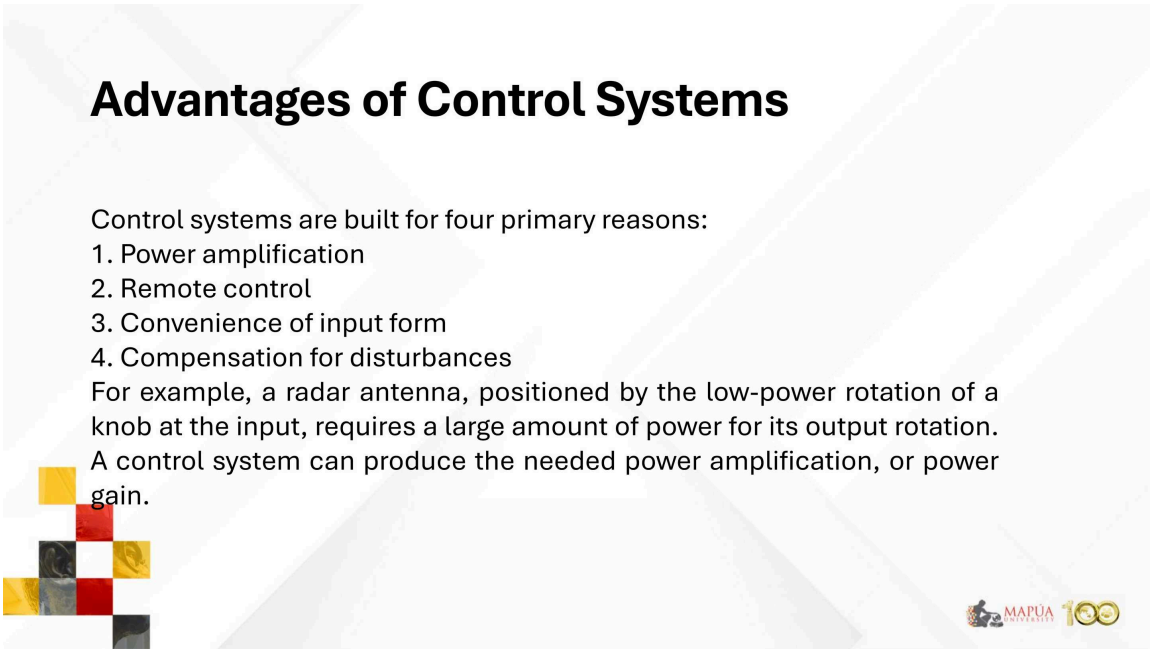
EXAMPLE: A part of the human temperature control system is the perspiration system. When the temperature of the air exterior to the skin becomes too high the sweat glands secrete heavily, inducing cooling of the skin by evaporation. Secretions are reduced when the desired cooling effect is achieved, or when the air temperature falls sufficiently.

The input to this system may be "normal" or comfortable skin temperature, a "setpoint," or the air temperature, a physical variable. The output is the actual skin temperature.



Advantages of Control Systems

With control systems we can move large equipment with precision that would otherwise be impossible. We can point huge antennas toward the farthest reaches of the universe to pick up faint radio signals; controlling these antennas by hand would be impossible. Because of control systems, elevators carry us quickly to our destination, automatically stopping at the right floor. We alone could not provide the power required for the load and the speed; motors provide the power, and control systems regulate the position and speed.



Advantages of Control Systems

Control systems are built for four primary reasons:

1. Power amplification
2. Remote control
3. Convenience of input form
4. Compensation for disturbances

For example, a radar antenna, positioned by the low-power rotation of a knob at the input, requires a large amount of power for its output rotation. A control system can produce the needed power amplification, or power gain.

Advantages of Control Systems

Robots designed by control system principles can compensate for human disabilities. Control systems are also useful in remote or dangerous locations. For example, a remote-controlled robot arm can be used to pick up material in a radioactive environment.
Control systems can also be used to provide convenience by changing the form of the input. For example, in a temperature control system, the input is a position on a thermostat. The output is heat. Thus, a convenient position input yields a desired thermal output.





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History of Control Systems

Liquid-Level Control
The Greeks began engineering feedback systems around 300 B.C. A water clock invented by Ktesibios operated by having water trickle into a measuring container at a constant rate. The level of water in the measuring container could be used to tell time. For water to trickle at a constant rate, the supply tank had to be kept at a constant level. This was accomplished using a float valve similar to the water-level control in today's flush toilets.





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History of Control Systems

Liquid-Level Control



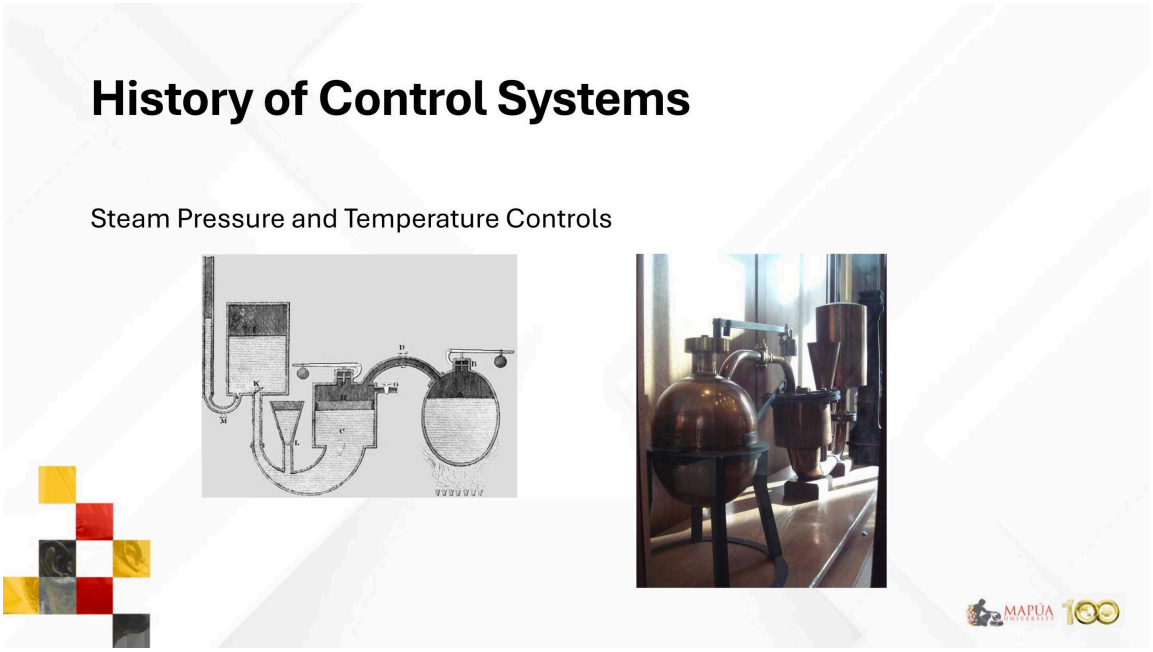
History of Control Systems

Steam Pressure and Temperature Controls
Regulation of steam pressure began around 1681 with Denis Papin's invention of the safety valve. The concept was further elaborated on by weighting the valve top. If the upward pressure from the boiler exceeded the weight, steam was released, and the pressure decreased. If it did not exceed the weight, the valve did not open, and the pressure inside the boiler increased. Thus, the weight on the valve top set the internal pressure of the boiler.

History of Control Systems

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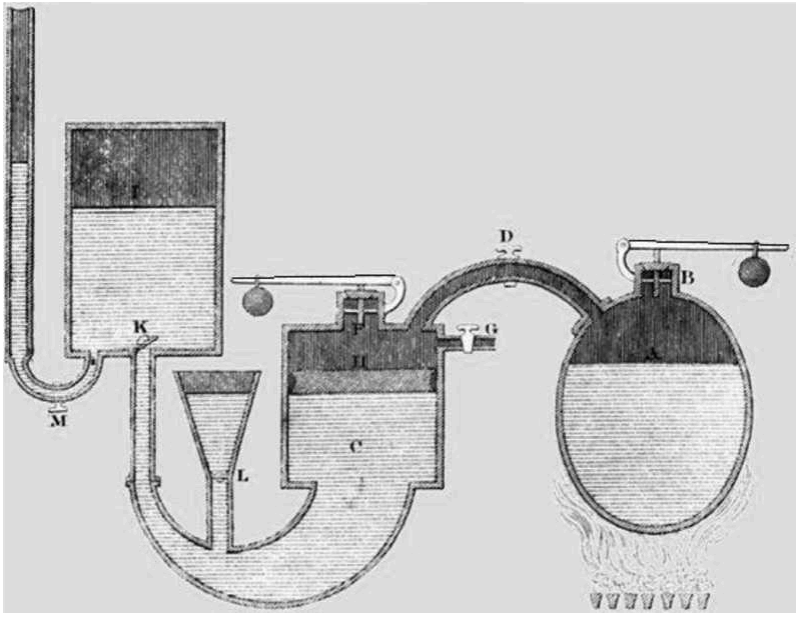
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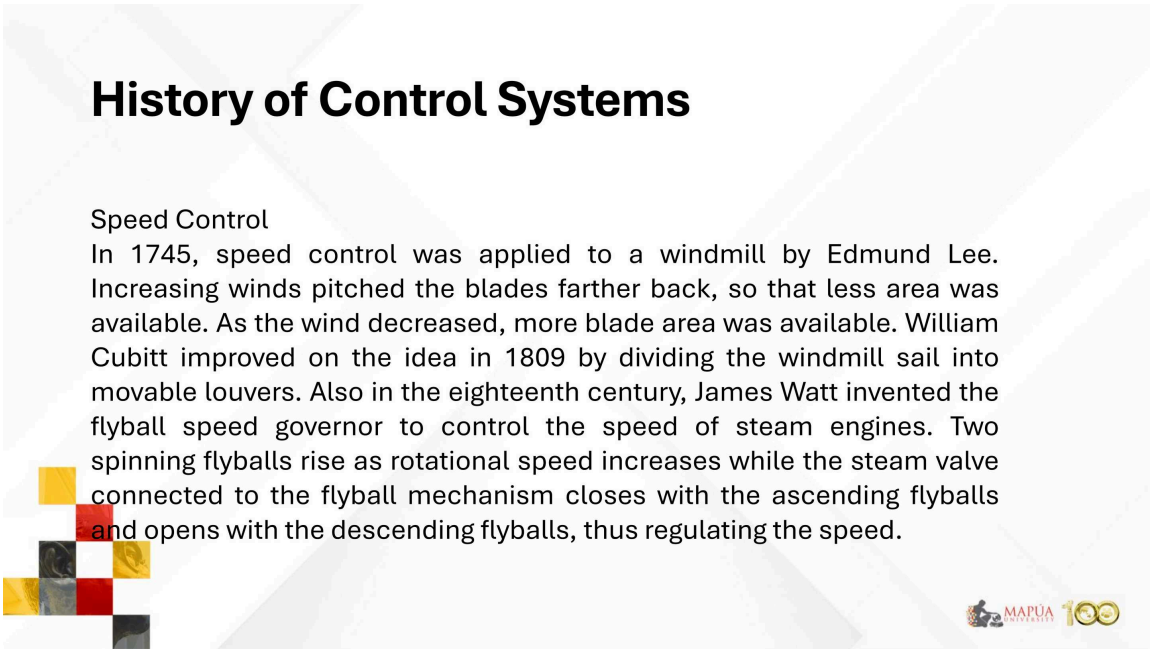


History of Control Systems

Steam Pressure and Temperature Controls




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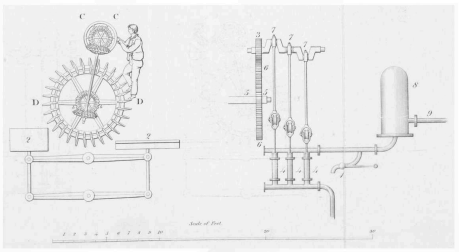
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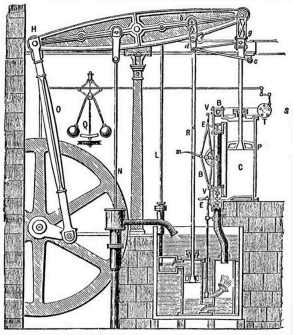
Speed Control

In 1745, speed control was applied to a windmill by Edmund Lee. Increasing winds pitched the blades farther back, so that less area was available. As the wind decreased, more blade area was available. William Cubitt improved on the idea in 1809 by dividing the windmill sail into movable louvers. Also in the eighteenth century, James Watt invented the flyball speed governor to control the speed of steam engines. Two spinning flyballs rise as rotational speed increases while the steam valve connected to the flyball mechanism closes with the ascending flyballs and opens with the descending flyballs, thus regulating the speed.

History of Control Systems

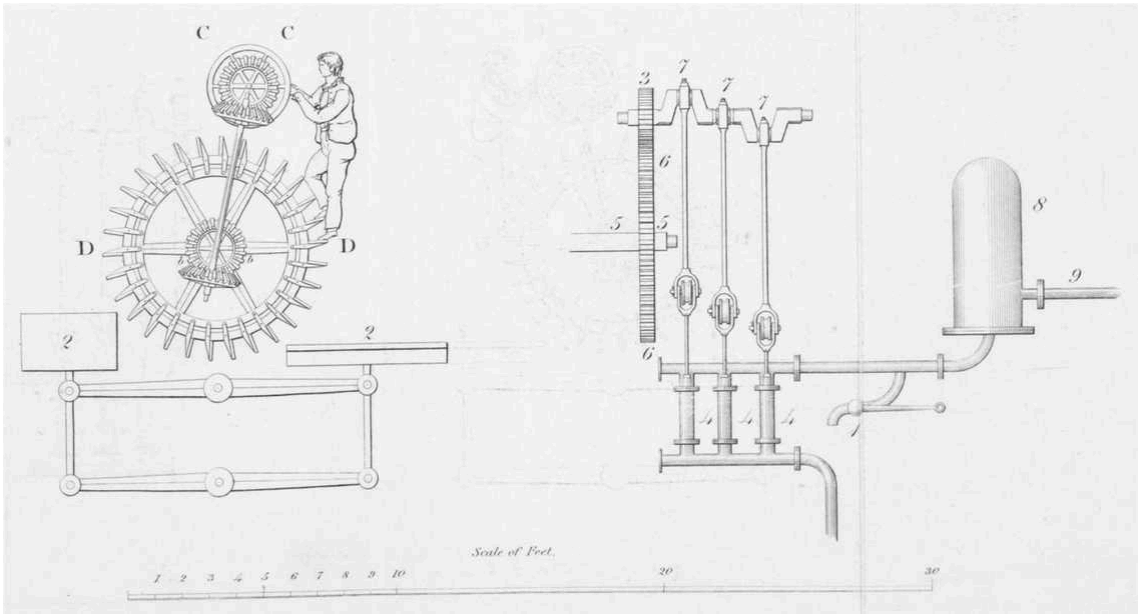
William Cubitt Treadwheel idea in 1809 and James Watt steam engine.

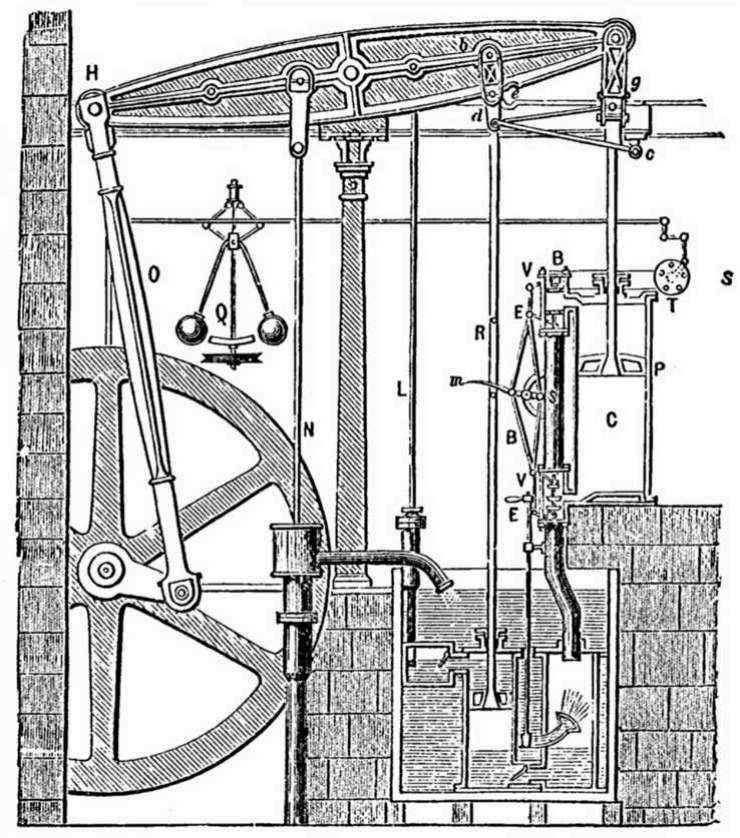




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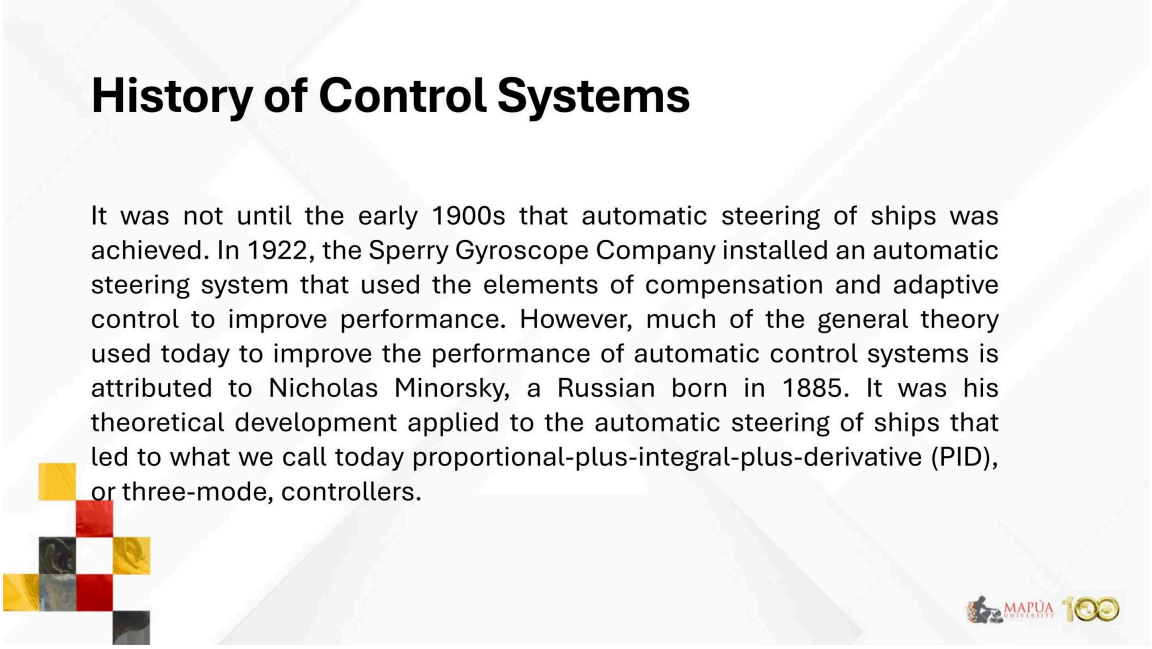
Stability, Stabilization, and Steering
Control systems theory as we know it today began to crystallize in the latter half of the nineteenth century. In 1868, James Clerk Maxwell published the stability criterion for a third-order system based on the coefficients of the differential equation. In 1874, Edward John Routh, using a suggestion from William Kingdon Clifford that was ignored earlier by Maxwell, was able to extend the stability criterion to fifth-order systems. In 1877, the topic for the Adams Prize was “The Criterion of Dynamical Stability.” In response, Routh submitted a paper entitled A Treatise on the Stability of a Given State of Motion and won the prize.

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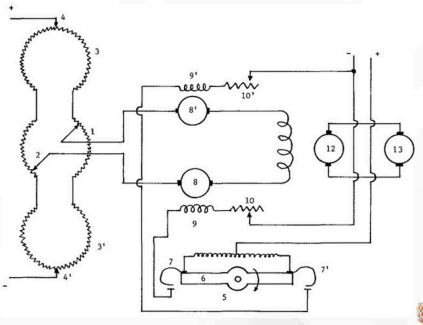
History of Control Systems

It was not until the early 1900s that automatic steering of ships was achieved. In 1922, the Sperry Gyroscope Company installed an automatic steering system that used the elements of compensation and adaptive control to improve performance. However, much of the general theory used today to improve the performance of automatic control systems is attributed to Nicholas Minorsky, a Russian born in 1885. It was his theoretical development applied to the automatic steering of ships that led to what we call today proportional-plus-integral-plus-derivative (PID), or three-mode, controllers.

History of Control Systems

Sperry Gyroscope Company automatic steering system and Nicholas Minorsky, theoretical proportional-plus-integral-plus-derivative (PID), or three-mode, controllers.

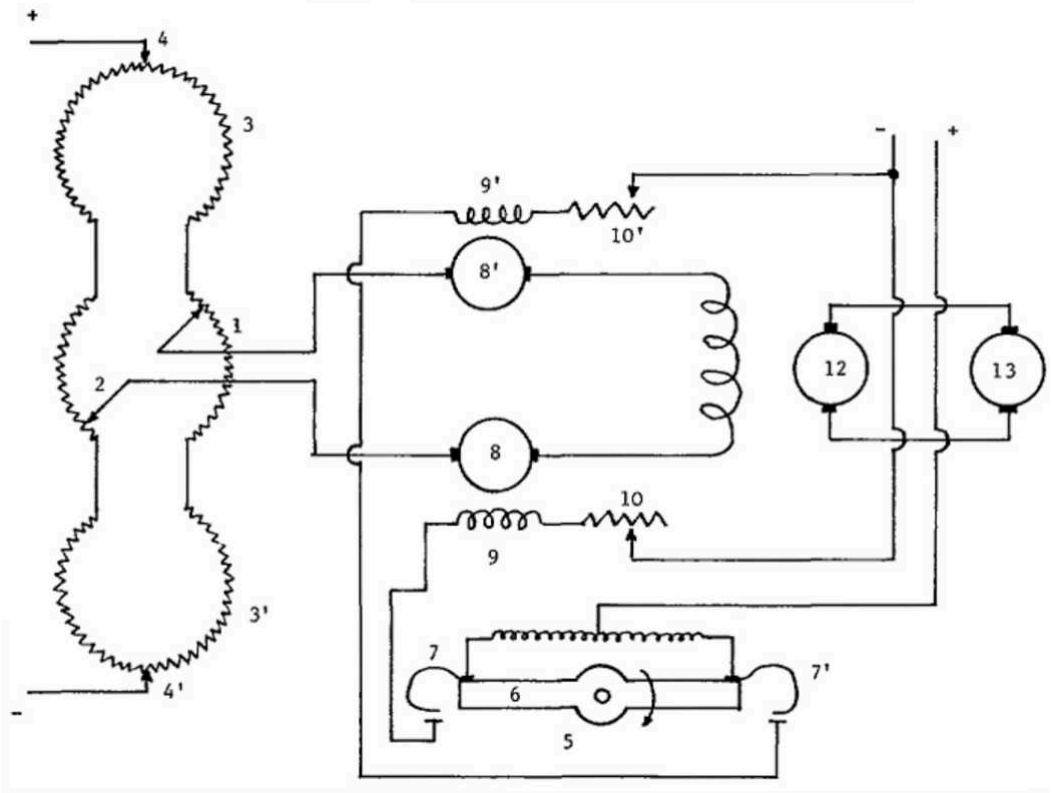




History of Control Systems

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System Configurations

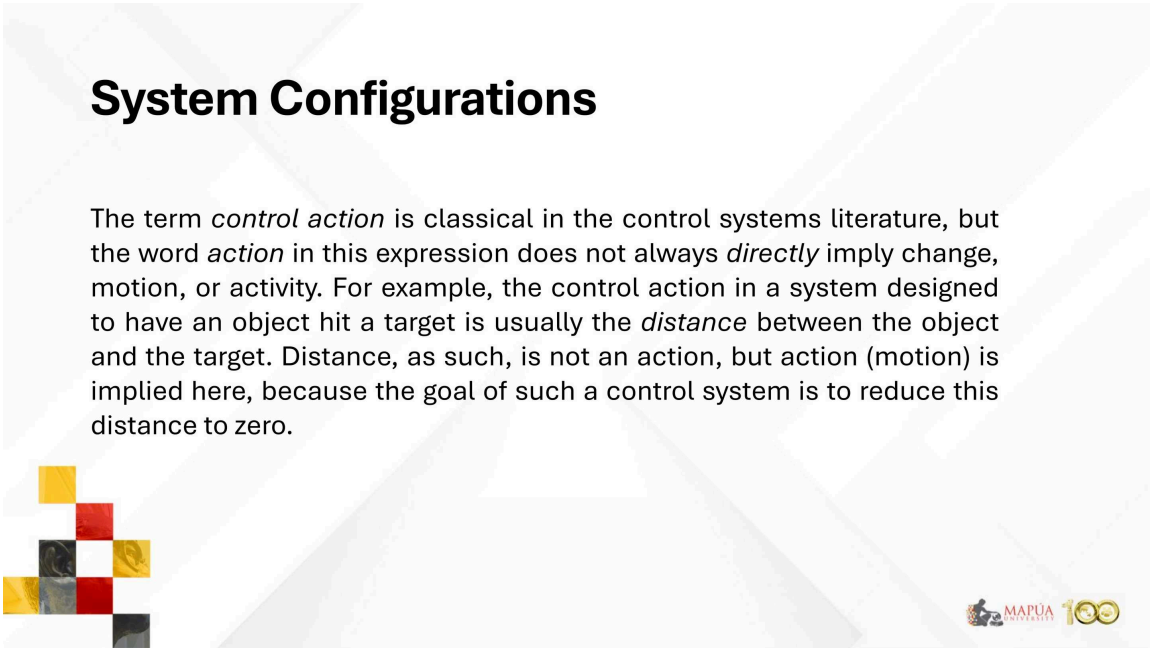
Control systems are classified into two general categories: *open-loop* and *closed-loop* systems. The distinction is determined by the **control action,** that quantity responsible for activating the system to produce the output. To classify a control system as open-loop or closed-loop, it is necessary to distinguish clearly the components of the system from components that interact with but are not part of the system.



System Configurations

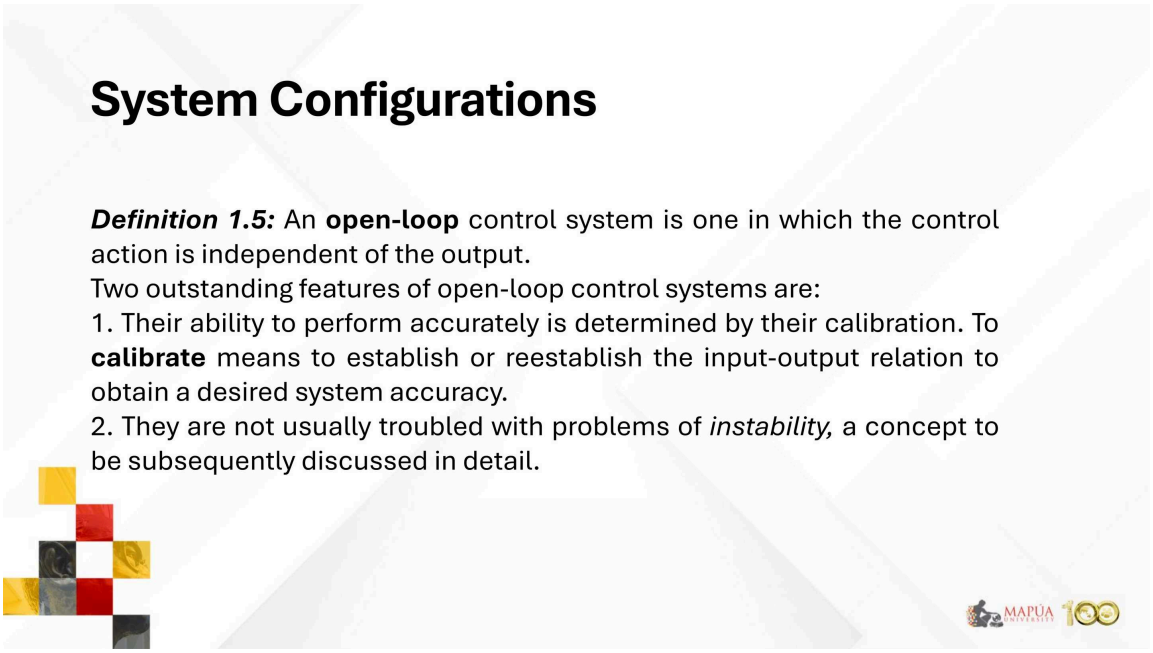
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System Configurations

The term control action is classical in the control systems literature, but the word action in this expression does not always directly imply change, motion, or activity. For example, the control action in a system designed to have an object hit a target is usually the distance between the object and the target. Distance, as such, is not an action, but action (motion) is implied here, because the goal of such a control system is to reduce this distance to zero.



System Configurations

Definition 1.5: An open-loop control system is one in which the control action is independent of the output.

Two outstanding features of open-loop control systems are:

1. Their ability to perform accurately is determined by their calibration. To calibrate means to establish or reestablish the input-output relation to obtain a desired system accuracy.
2. They are not usually troubled with problems of instability, a concept to be subsequently discussed in detail.

System Configurations

EXAMPLE: Most *automatic toasters* are open-loop systems because they are controlled by a timer. The time required to make "good toast" must be estimated by the user, who is not part of the system. Control over the quality of toast (the output) is removed once the time, which is both the input and the control action, has been set. The time is typically set by means of a calibrated dial or switch.



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System Configurations

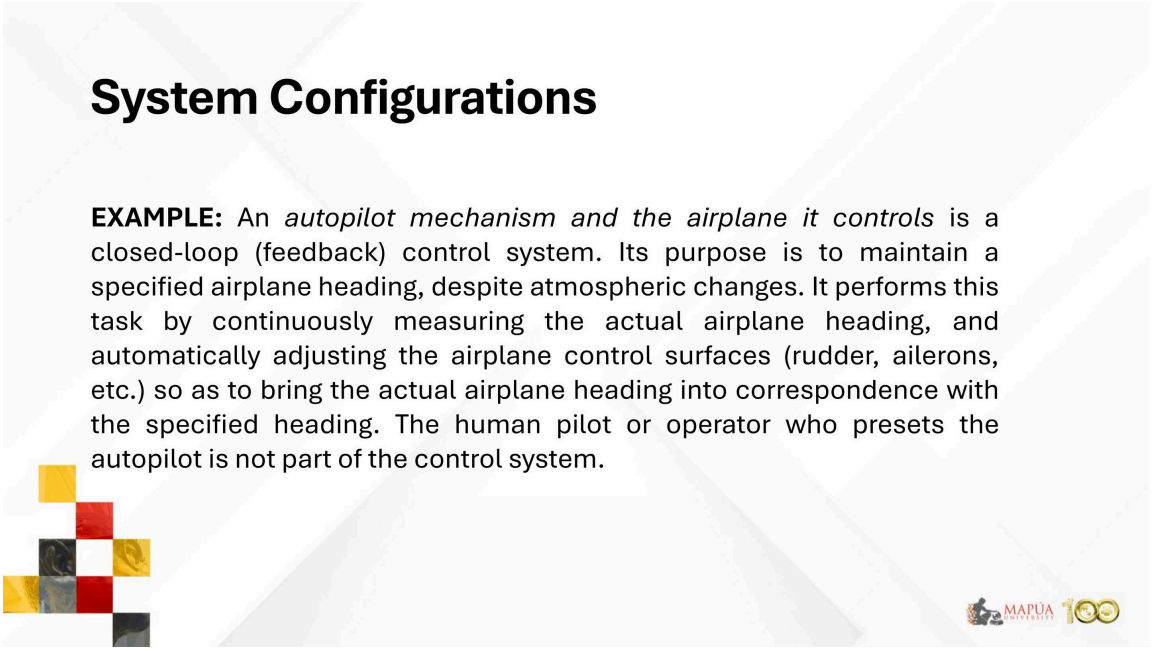
Definition 1.6: *A* **closed-loop** control system is one in which the control action is somehow dependent on the output. Closed-loop control systems are more commonly called *feedback* control systems.
Feedback is that characteristic of closed-loop control systems which distinguishes them from open-loop systems. feedback is said to exist in a system when a *closed* sequence of cause-and-effect relations exists between system variables.



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System Configurations

EXAMPLE: An autopilot mechanism and the airplane it controls is a closed-loop (feedback) control system. Its purpose is to maintain a specified airplane heading, despite atmospheric changes. It performs this task by continuously measuring the actual airplane heading, and automatically adjusting the airplane control surfaces (rudder, ailerons, etc.) so as to bring the actual airplane heading into correspondence with the specified heading. The human pilot or operator who presets the autopilot is not part of the control system.

System Configurations

Definition 1.7: **Feedback** is that property of a closed-loop system which permits the output (or some other controlled variable) to be compared with the input to the system (or an input to some other internally situated component or subsystem) so that the appropriate control action may be formed as some function of the output and input.



System Configurations

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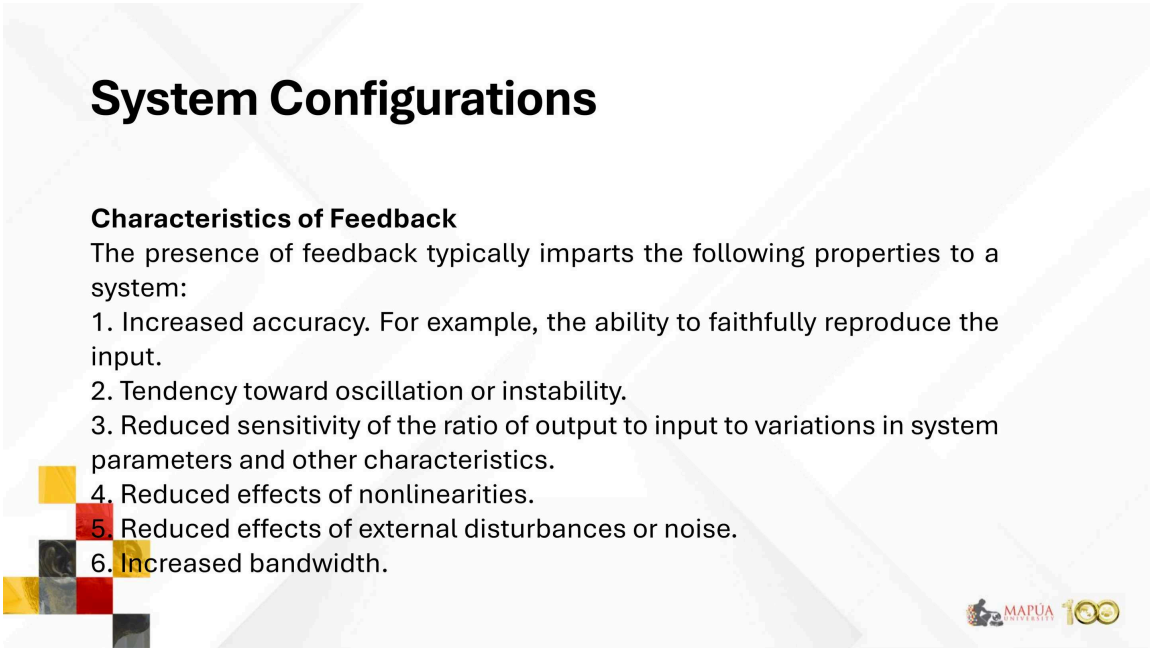
System Configurations

EXAMPLE: The concept of feedback is clearly illustrated by the autopilot mechanism. The input is the specified heading, which may be set on a dial or other instrument of the airplane control panel, and the output is the actual heading, as determined by automatic navigation instruments. A comparison device continuously monitors the input and output. When a difference exists between the input and output, the comparison device delivers a control action signal to the controller, the autopilot mechanism. Feedback may be effected by mechanical or electrical connections from the navigation instruments, measuring the heading, to the comparison device.



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System Configurations

Characteristics of Feedback

The presence of feedback typically imparts the following properties to a system:

1. Increased accuracy. For example, the ability to faithfully reproduce the input.
2. Tendency toward oscillation or instability.
3. Reduced sensitivity of the ratio of output to input to variations in system parameters and other characteristics.
4. Reduced effects of nonlinearities.
5. Reduced effects of external disturbances or noise.
6. Increased bandwidth.

Analog and Digital Control Systems

Definition 1.8: A signal dependent on a continuum of values of the independent variable *t* is called a **continuous-time** signal or, more generally, a **continuous-data** signal or (less frequently) an **analog** signal.
Definition 1.9: A signal defined at, or of interest at, only discrete (distinct) instants of the independent variable / (upon which it depends) is called a **discrete-time,** a **discrete data,** a **sampled-data,** or a **digital** signal.



Analog and Digital Control Systems

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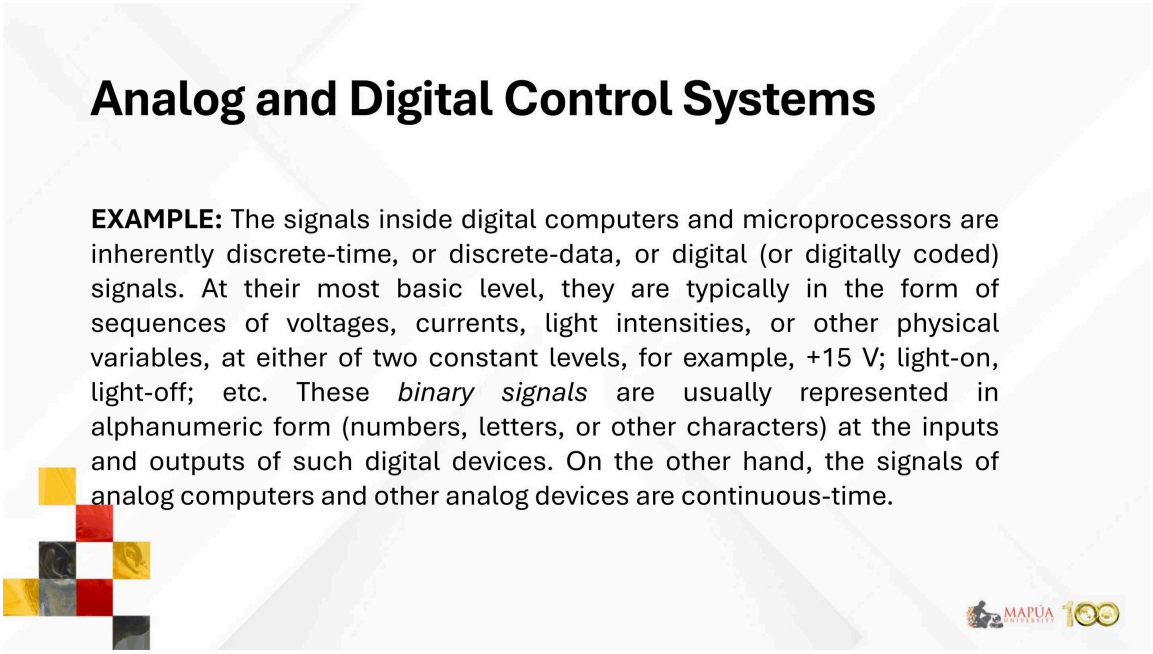
Analog and Digital Control Systems

EXAMPLE: The continuous, sinusoidally varying voltage *v(t)* or alternating current *i(t)* available from an ordinary household electrical receptacle is a continuous-time (analog) signal, because it is defined at *each and every instant* of time *t* electrical power is available from that outlet.



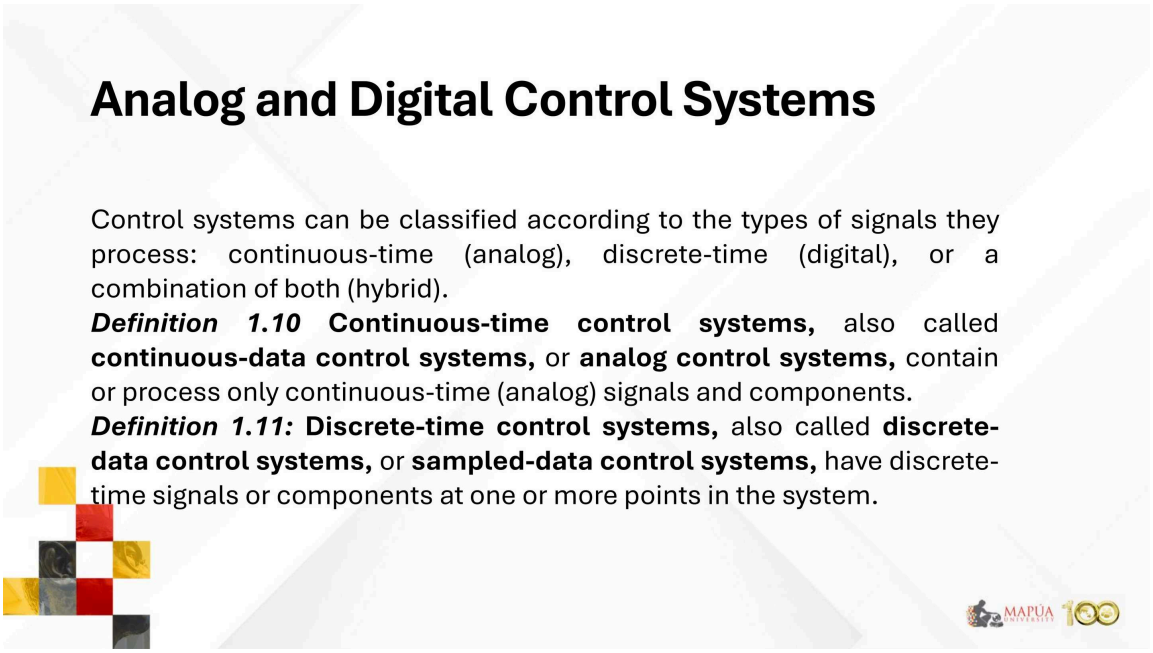
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Analog and Digital Control Systems

EXAMPLE: The signals inside digital computers and microprocessors are inherently discrete-time, or discrete-data, or digital (or digitally coded) signals. At their most basic level, they are typically in the form of sequences of voltages, currents, light intensities, or other physical variables, at either of two constant levels, for example, +15 V; light-on, light-off; etc. These binary signals are usually represented in alphanumeric form (numbers, letters, or other characters) at the inputs and outputs of such digital devices. On the other hand, the signals of analog computers and other analog devices are continuous-time.



Analog and Digital Control Systems

Control systems can be classified according to the types of signals they process: continuous-time (analog), discrete-time (digital), or a combination of both (hybrid).

Definition 1.10 Continuous-time control systems, also called continuous-data control systems, or analog control systems, contain or process only continuous-time (analog) signals and components.

Definition 1.11: Discrete-time control systems, also called discrete-data control systems, or sampled-data control systems, have discrete-time signals or components at one or more points in the system.

Analog and Digital Control Systems

EXAMPLE: A target tracking and following system, (tracking and pointing at an object with a finger), is usually considered an analog or continuous-time control system, because the distance between the "tracker" (finger) and the target is a continuous function of time, and the objective of such a control system is to *continuously* follow the target. The system consisting of a person driving an automobile falls in the same category.



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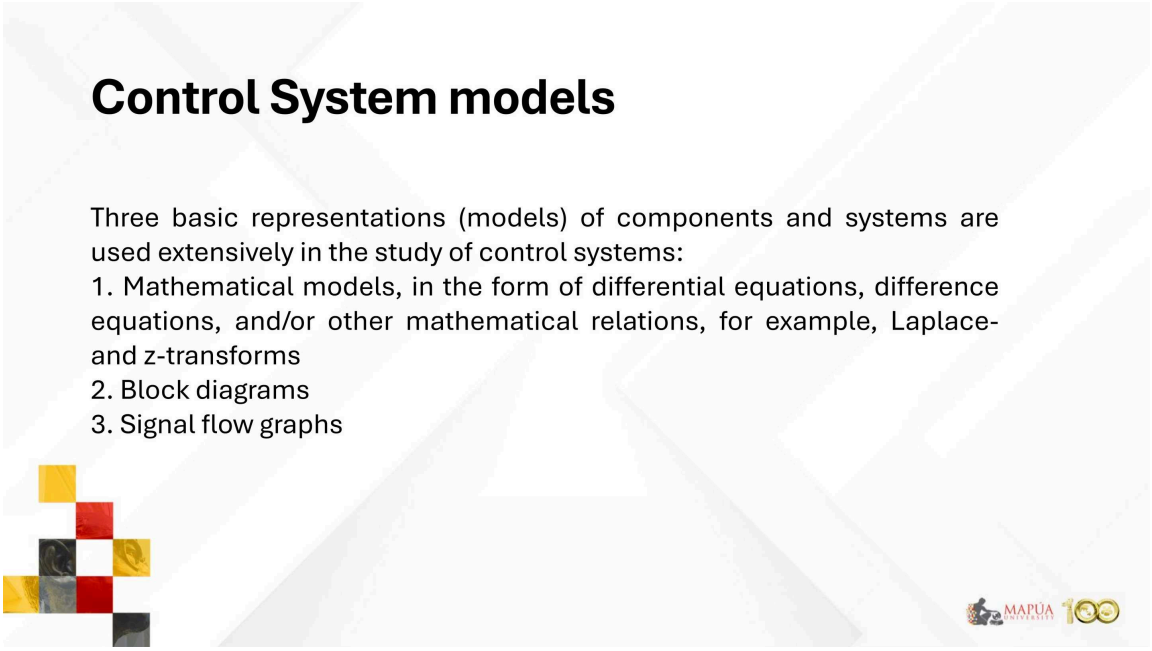
Analog and Digital Control Systems

EXAMPLE: A closer look at the thermostatically controlled heating system of Example 1.2 indicates that it is actually a sampled-data control system, with both digital and analog components and signals. If the desired room temperature is, say, 68°F (22°C) on the thermostat and the room temperature falls below, say, 66°F, the thermostat switching system closes the circuit to the furnace (an analog device), turning it on until the temperature of the room reaches, say, 70°F. Then the switching system automatically turns the furnace off until the room temperature again falls below 66°F.



Analog and Digital Control Systems

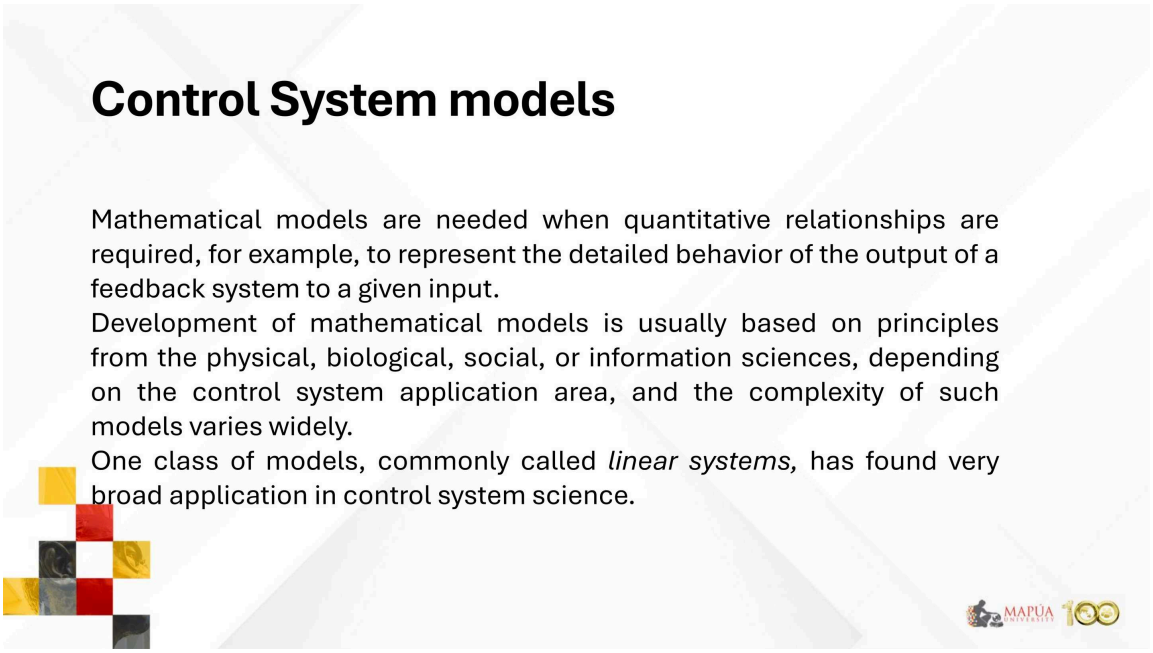
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Control System models

Three basic representations (models) of components and systems are used extensively in the study of control systems:

1. Mathematical models, in the form of differential equations, difference equations, and/or other mathematical relations, for example, Laplace- and z-transforms
2. Block diagrams
3. Signal flow graphs



Control System models

Mathematical models are needed when quantitative relationships are required, for example, to represent the detailed behavior of the output of a feedback system to a given input.

Development of mathematical models is usually based on principles from the physical, biological, social, or information sciences, depending on the control system application area, and the complexity of such models varies widely.

One class of models, commonly called linear systems, has found very broad application in control system science.



Thank you!